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Effective Suppression of Lithium Dendrite Growth Using a Flexible Single-Ion Conducting Polymer Electrolyte

Kuirong Deng, Jiaxiang Qin, Shuanjin Wang, Shan Ren, Dongmei Han, Min Xiao, and Yuezhong Meng**

A novel single-ion conducting polymer electrolyte (SIPE) membrane with high lithium-ion transference number, good mechanical strength, and excellent ionic conductivity is designed and synthesized by facile coupling of lithium bis(allylmalonato) borate (LiBAMB), pentaerythritol tetrakis (2-mercaptoacetate) (PETMP) and 3,6-dioxa-1,8-octanedithiol (DODT) in an electrospun poly(vinylidienefluoride) (PVDF) supporting membrane via a one-step photo-initiated in situ thiol–ene click reaction. The structure-optimized LiBAMB-PETMP-DODT (LPD)@PVDF SIPE shows an outstanding ionic conductivity of 1.32 × 10^{-3} S cm^{-1} at 25 °C, together with a high lithium-ion transference number of 0.92 and wide electrochemical window up to 6.0 V. The SIPE exhibits high tensile strength of 7.2 MPa and elongation at break of 269%. Due to these superior performances, the SIPE can suppress lithium dendrite growth, which is confirmed by galvanostatic Li plating/stripping cycling test and analysis of morphology of Li metal electrode surface after cycling test. Li|LPD@PVDF|Li symmetric cell maintains an extremely stable and low overpotential without short circuiting over the 1050 h cycle. The Li|LPD@PVDF|$LiFePO_4$ cell shows excellent rate capacity and outstanding cycle performance compared to cells based on a conventional liquid electrolyte (LE) with Celgard separator. The facile approach of the SIPE provides an effective and promising electrolyte for safe, long-life, and high-rate lithium metal batteries.

1. Introduction

Lithium-ion batteries (LIBs) are widely applied in consumer electronics, smart grids, electric vehicles (EVs), and hybrid electric vehicles (HEVs). However, they cannot meet the demand for higher power density, enhanced safety, and lower cost in applications such as electrical vehicles and autonomous aircrafts.[1] Lithium metal batteries (LMBs), which use lithium metal instead of graphite as the anode, are attractive alternative to conventional LIBs for their much higher energy density.[2] Since lithium metal can offer the highest theoretical specific capacity of 3860 mA h g^{-1} and the lowest redox potential (−3.04 V vs the standard hydrogen electrode).[2b] What's more, lithium metal can play the role of lithium source, enabling the use of high capacity, unlithiated cathode such as sulfur and oxygen as well as the elimination of the current collectors used in conventional anodes.[3] However, lithium metal is highly reactive with flammable and volatile organic electrolytes, leading to uneven lithium deposition and lithium dendrite formation during charge/discharge cycles, which thereby cause short cycle life, low coulombic efficiency, and serious safety issues including internal short circuits and thermal runaway of the batteries.[4] Such serious problems prevent the practical application of LMBs.

Over the last few decades, extensive research has been devoted to suppress lithium dendrite growth, including ex situ protective coating of lithium anode,[5] using additives to tune the solid electrolyte interphase in liquid electrolytes[4c,6] and utilization of solid polymer electrolytes (SPEs).[7] So far, two main theoretical frameworks have been proposed to understanding suppression of lithium dendrite growth. Chazalviel et al. suggested that the anion depletion near Li electrode could lead to large electric fields, which in turn causes dendrite growth.[8] Thus, electrolytes with low anion mobility (i.e., high lithium-ion transference number) and high ionic conductivity can suppress lithium dendrite growth by preventing large electric fields.[2b] The other model proposed by Monroe and Newman suggests that lithium dendrite growth can be mechanically blocked if the shear modulus of the electrolytes is about twice that of lithium metal.[9] Based on the combination of the two theoretical frameworks, utilization of electrolytes with high lithium-ion transference number and high modulus is an effective approach.[10] Although SPEs are able to suppress lithium dendrite growth, ionic conductivity of SPEs at room temperature is too low (10^{-7}–10^{-5} S cm^{-1}).[7a,11] In general, gel polymer electrolytes (GPEs) show higher ionic conductivity than SPE. Nevertheless, GPEs usually suffer from poor mechanical strength and low lithium-ion transference number.[12]

K. Deng, Dr. J. Qin, Prof. S. Wang, Prof. S. Ren, Dr. D. Han, Prof. M. Xiao, Prof. Y. Meng
The Key Laboratory of Low-Carbon Chemistry and Energy Conservation of Guangdong Province/State Key Laboratory of Optoelectronic Materials and Technologies
School of Materials Science and Engineering
Sun Yat-sen University
Guangzhou 510275, P. R. China
E-mail: stsxm@mail.sysu.edu.cn; mengyzh@mail.sysu.edu.cn

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To achieve high lithium-ion transference number, single-ion conducting polymer electrolytes (SIPEs) were proposed, within which the anions are covalently immobilized on rigid polymer chains and the counter lithium ions are the dominant transporting species.[13] There are many reports on solid-state SIPEs[7b,14] and gel-state SIPEs[15] with enhanced lithium-ion transference number around 0.9. poly(methylmethacrylate) (PMMA)-IL-TFSI/LiTFSI gel-state SIPEs based on ionic liquid (IL)-decorated PMMA nanoparticles dispersed in a propylene carbonate (PC) were reported to be effective in suppressing lithium dendrite formation.[15b] But current density applied for galvanostatic Li plating/stripping cycling tests is only 0.2 mA cm^{-2} and corresponding overpotential is as high as 0.18 V. Except for this report, there have been few discussions about the ability of SIPEs to suppress lithium dendrite growth reported. Further, most of these reported gel-state SIPEs are prepared by blending poly lithium salt with linear poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP). When plasticized by organic solvents, the resulting gel SIPEs easily lose mechanical strength.[16] To improve mechanical strength and dimensional stability, it is an effective method to incorporate chemically cross-linked structure interpenetrating the host polymer skeleton.[17]

In this work, we report a novel gel-state SIPE with high lithium-ion transference number, good mechanical strength, and excellent ionic conductivity prepared by facile and green method. A three dimensional (3D) network structure was formed within the electrospun PVDF membranes, via one-step photoinitiated in situ thiol–ene click reaction of LiBAMB, pentaerythritol tetrakis(2-mercaptoacetate) (PETMP), and DODT in the presence of gamma-butyrolactone (GBL). In this composite structure of LPD@PVDF, electrospun PVDF acts as the supporting framework to enhance the mechanical strength of the polymer network. The anions $BAMB^-$ are covalently tethered into the polymer network to restrict the anionic movement, which result in high lithium-ion transference number (0.92). Charge delocalization of boron atoms in $BAMB^-$ is achieved by covalently bonded with electron-withdrawing groups, which lead to weak electrostatic interaction between lithium ions and the anions. Thus, the gel SIPE shows enhanced ionic conductivity (>10^{-3} S cm^{-1} at 25 °C). Due to high lithium-ion transference number and good mechanical strength of LPD@PVDF SIPE, it is able to suppress lithium dendrite growth. Li|LPD@PVDF|Li symmetric cell maintains an extremely stable and low overpotential without short circuiting over 1050 h cycle. Li|LPD@PVDF|$LiFePO_4$ cell shows excellent rate capacity and cycle performance. It can keep a steady discharge capacity of 112 mA h g^{-1} at 5 C and maintain discharge capacity of 128 mA h g^{-1} at 1 C rate after 380 cycles, which is 91% retention of the initial capacity.

2. Results and Discussion

2.1. Preparation Procedure of LPD@PVDF SIPEs

LPD@PVDF SIPEs were prepared by facile and green method as shown in **Figure 1**. Electrospun PVDF membranes were immersed into the precursor and exposed to ultraviolet (UV) light (365 nm). Photoinitiated in situ thiol–ene click reaction would take place immediately, which is characteristic for high reaction selectivity, high conversion, fast reaction rate, and mild reaction conditions.[18] A 3D network was constructed within the electrospun PVDF membranes by coupling of LiBAMB, PETMP, and DODT via one-step thiol–ene click reaction. The prepared LPD@PVDF membranes are flexible and transparent as shown in **Figure 2**a and Video S1 (Supporting Information). The surface morphology of electrospun PVDF membrane was characterized by scanning electron microscopy (SEM), as shown in Figure 2b. The fibers appear to be uniform and interlaid to form a 3D network with a large number of micrometer and sub-micrometer-sized interconnected pores in the structure. While for LPD@PVDF SIPE membrane (Figure 2c), the pores are smoothly filled with gel. Sulfur is distributed uniformly as shown in energy dispersive spectrometer (EDS) mapping of sulfur (Figure 2d), indicating that distribution of crosslinking LPD is uniform.

2.2. Ionic Conductivity

Due to 3D cross-linked structure of LPD in LPD@PVDF SIPEs, crosslinking density may influence the movement of the polymer chain and thus influence lithium-ion transport. To investigate the influence of crosslinking density on ionic conductivity, LPD (without electrospun PVDF) was in situ formed in measurement device (Figure S1, Supporting Information). In LPD SIPEs, PETMP is the only cross-link point. So molar fraction of PETMP represents crosslinking density. As shown in **Figure 3**a, when PETMP molar fraction increases from 11.1 to 33.3%, ionic conductivity decreases from 1.99×10^{-3} to 1.73×10^{-3} S cm^{-1}. It is ascribed to the increased crosslinking density of the 3D network. Increased crosslinking density hinders the mobility of the lithium ion in the polymer matrix. LPD SIPEs with 20.0% PETMP show improved ionic conductivity compared to that of LPD SIPEs with 33.3% PETMP and better mechanical strength than that of LPD SIPEs with 14.3% PETMP. So PETMP molar fraction of 20.0% was chosen as the optimal ratio.

Influence of precursor concentration on ionic conductivity of LPD@PVDF SIPEs with 20.0% PETMP was investigated. Electrospun PVDF membranes were immersed into precursors with various concentrations and cured under UV light (365 nm). Ionic conductivity of these LPD@PVDF SIPEs is shown in Figure 3b. When concentration of LiBAMB increases from 0.25 to 0.33 mol L^{-1}, ionic conductivity increases from 1.18×10^{-3} to 1.32×10^{-3} S cm^{-1}. Such behavior can be reasonably explained by the increment in concentration of lithium ion in the electrolyte. While when concentration of LiBAMB increases from 0.33 to 0.66 mol L^{-1}, ionic conductivity decreases from 1.32×10^{-3} to 0.89×10^{-3} S cm^{-1}, which is attributed to the decreased fraction of plasticizer within LPD@PVDF SIPEs. Lower fraction of plasticizer in gel is unfavorable for the movement of lithium ion in the polymer matrix, which causes reduction in ionic conductivity.

Under the same conditions (precursor with 20.0% PETMP and concentration of 0.33 mol L^{-1}), the ionic conductivity of LPD (without electrospun PVDF) is 1.89×10^{-3} S cm^{-1}. However,

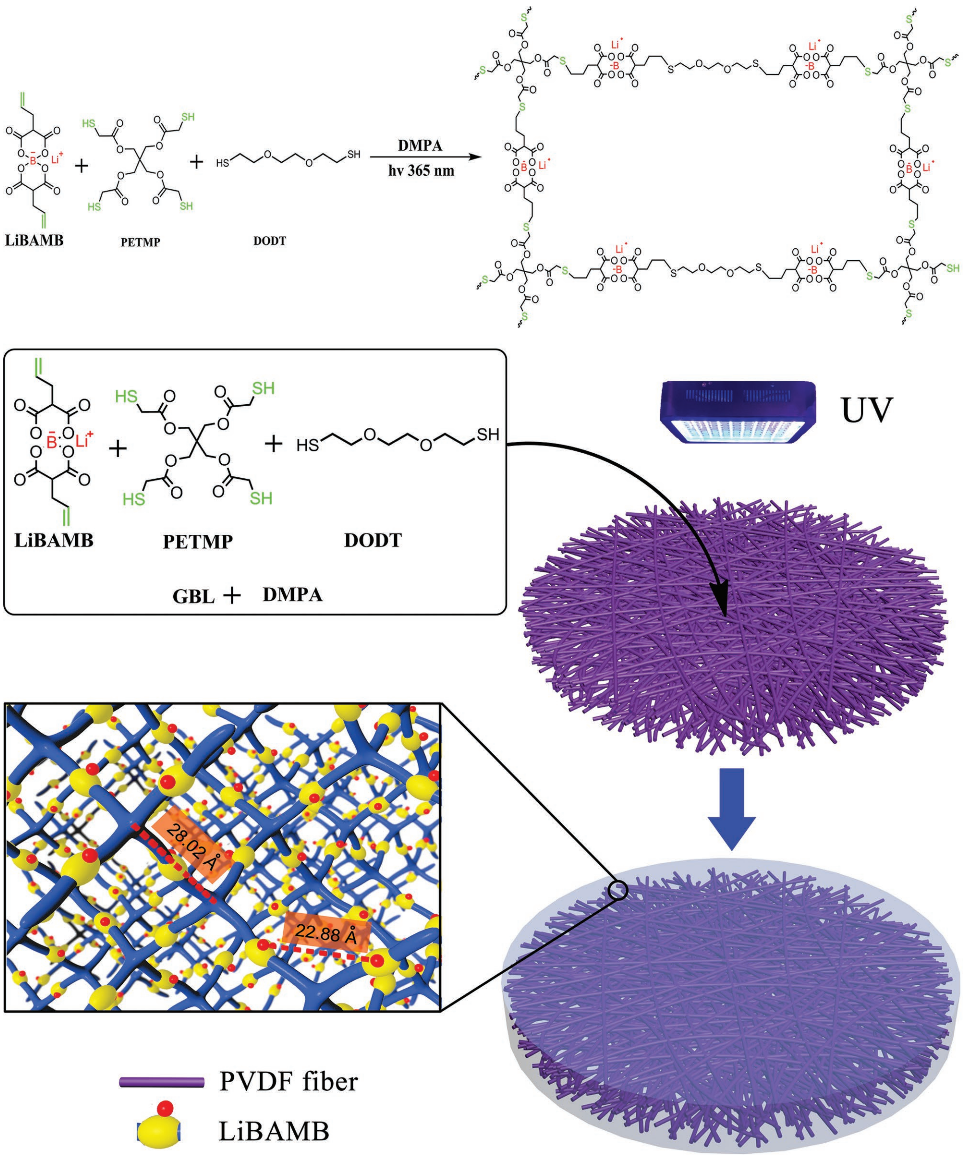

Figure 1. Schematic illustration of the synthetic process and structure of LPD@PVDF SIPEs.

with electrospun PVDF, the ionic conductivity of LPD@PVDF decreased slightly to 1.32×10^{-3} S cm^{-1}. Although the electrospun PVDF fibers appear to be a 3D network structure with a large number of micrometer and sub-micrometer sized interconnected pores, lithium ion can transport through these interconnected pores. Nevertheless, the PVDF fibers will decrease the content of LPD and hinder movement of lithium ion to some content.

Temperature dependence of ionic conductivity for LPD@PVDF SIPEs made from precursor at concentration of 0.33 mol L^{-1} (LiBAMB) is measured from room temperature to 80 °C with an increment of 10 °C and drawn as dots in Figure 3c. The ionic conductivity increases with increasing temperature. At 80 °C, ionic conductivity reaches 2.66×10^{-3} S cm^{-1}. Fits to the Arrhenius equation are shown in Figure 3c as lines. Ionic conductivity of LPD@PVDF SIPEs follows obviously an Arrhenius behavior. LPD@PVDF SIPEs made from precursor with concentration of 0.33 mol L^{-1} (LiBAMB) show the highest ionic conductivity of 1.32×10^{-3} S cm^{-1} at 25 °C, which is among the highest ionic conductivity exhibited for single-ion conducting electrolytes reported to date as summarized in Table S1 (Supporting Information). This ionic conductivity is also higher than that of combination of polyolefin separators and liquid electrolytes, generally around 0.1–1 × 10^{-3} S cm^{-1}.[15a] For example, ionic conductivity of Celgard 2500 separator combined with 1 M $LiPF_6$ in DMC + EC was only 0.85×10^{-3} S cm^{-1}.[19] Such high ionic conductivity is ascribed to charge delocalization of boron atoms in BAMB$^-$ which was achieved by covalently bonded with electron-withdrawing groups. Electrostatic interaction between lithium ions and the anions is weak. Furthermore, dissociation and mobility of lithium ion is enhanced by appropriate plasticizer GBL.

Density functional theory (DFT) calculations were performed to investigate the dissociation of LiBAMB and

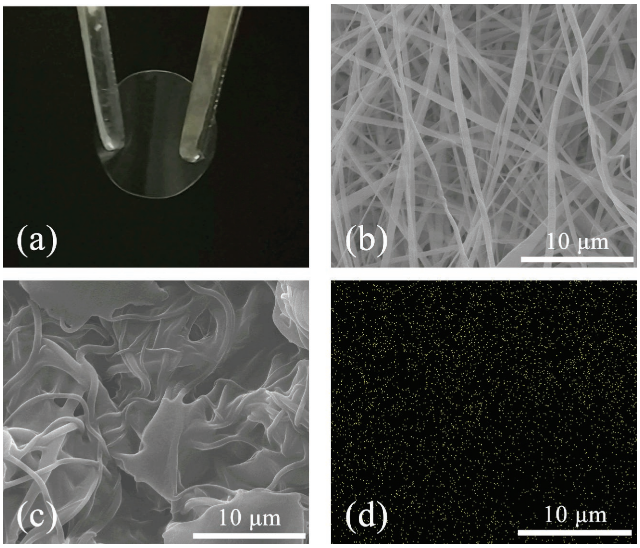

Figure 2. a) Photograph of LPD@PVDF SIPE membrane. b) SEM image of electrospun PVDF membrane. c) SEM image of LPD@PVDF SIPE membrane. d) EDS mapping of sulfur corresponding to (c).

solvation of lithium ions in LPD@PVDF SIPE. The optimized structures of the LiBAMB and $Li^+(GBL)_4$ solvation complex are shown in Figure 3d and their energy are listed. The bond length of Li—B is 3.233 Å and the binding energy of LiBAMB is low as −143.0 kcal mol^{-1}, indicating weak electrostatic interaction between lithium ions and the anions. In the $Li^+(GBL)_n$ solvation complexes, O atoms of the carbonyl group attach by coordination to the Li ions. $Li^+(GBL)_4$ show the lowest solvation energy of −151.7 kcal mol^{-1}. It is clear that dissociation energy of LiBAMB is lower than solvation energy of $Li^+(GBL)_4$, so lithium ions tend to dissociate from LiBAMB and form solvation structure $Li^+(GBL)_4$. Consequently, the most stable solvation structures are $Li^+(GBL)_4$ and $Li^+(GBL)_4$ is the main solvation mode of lithium ions, which is well agreed with previous report.[20] Weak electrostatic interaction between lithium ions and the anions and stable solvation structure $Li^+(GBL)_4$ contribute to high ionic conductivity LPD@PVDF SIPE.

2.3. Lithium-Ion Transference Number

The transference number was evaluated via combination of chronoamperometry (CA) and electrochemical impedance spectroscopy (EIS) before and after the CA test, which is based on the method proposed by Evans et al.[21] Chronoamperometry and electrochemical impedance spectroscopy before and after the CA test are shown in **Figure 4**a,b. It was observed that the current response to the applied static potential polarization drops initially and then remains steady for the rest of measurement. The impedance increased a little after the chronoamperometry measurement. The lithium-ion transference number is calculated to be 0.92, close to unity, indicating a single-ion conducting behavior. The significantly high value of T_+ value is due to the restricted movement of anions as the part of crosslinking network.

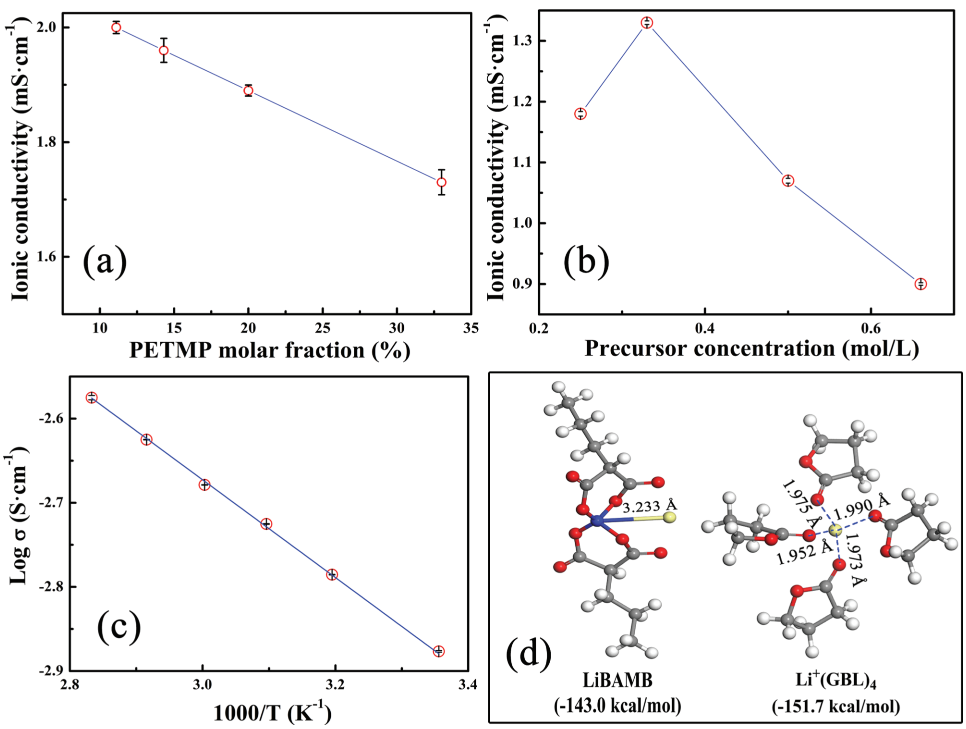

Figure 3. a) Influence of PETMP molar fraction on ionic conductivity of LPD at 25 °C; b) influence of precursor concentration on ionic conductivity of LPD@PVDF SIPEs at 25 °C; c) temperature dependence and Arrhenius fitting of ionic conductivity (precursor concentration 0.33 mol L^{-1}); d) models of optimum structure of LiBAMB and $Li^+(GBL)_4$ and related complexation energies.

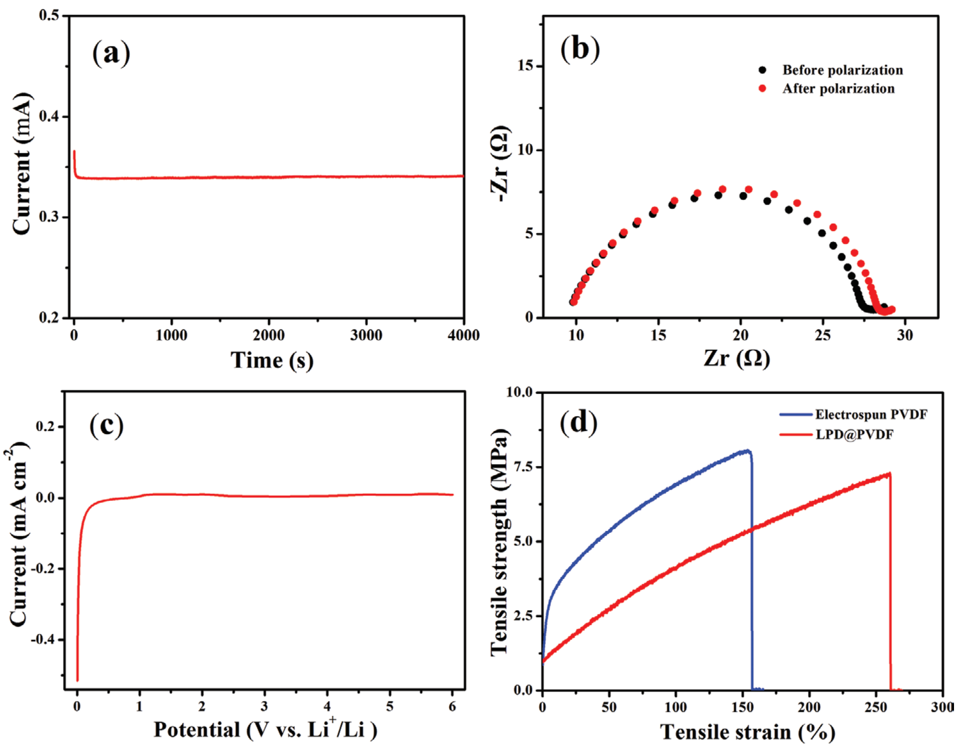

Figure 4. a) Chronoamperometry of Li|LPD@PVDF|Li symmetric cell at a static potential of 10 mV; b) Nyquist plots of the Li|LPD@PVDF|Li symmetric cell before/after polarization; c) linear sweep voltammetry of the Li|LPD@PVDF|SS cell from 0 to 6.0 V (vs Li^+/Li) at a scan rate of 1 mV s^{-1}; d) stress–strain curves of electrospun PVDF and LPD@PVDF membrane.

2.4. Electrochemical Stability

The electrochemical stability of LPD@PVDF SIPEs was analyzed by linear sweep voltammetry (LSV). LSV was conducted from 0 to 6.0 V (vs Li^+/Li) at a scan rate of 1 mV s^{-1}. As shown in Figure 4c, there is no significant increase in the oxidation current up to 6.0 V, indicating a superior broad electrochemical window of 6.0 V. This superior electrochemical stability suggests that LPD@PVDF SIPEs can be applied to most commonly batteries, such as $LiFePO_4$ (3.8 V) and $LiNi_{0.5}Mn_{1.5}O_4$ (4.7 V).[22]

2.5. Mechanical Property

The mechanical property of LPD@PVDF membrane was evaluated by performing tensile strength testing. It is important to note that the tested membrane was in the same state as applied in batteries (gel state). However, most gel SIPEs reported were tested in dry state, which cannot reflect the real mechanical property as applied in batteries.[15d–i] As shown in Figure 4d, the tensile strength of the electrospun PVDF is 8.0 MPa with an elongation at break of 165%. While LPD@PVDF membrane shows a tensile strength of 7.2 MPa and a higher elongation at break of 269%, which demonstrates its sufficient high mechanical strength to act the dual role of an electrolyte as well as a separator during the batteries fabrication and operation. Good mechanical strength is attributed to electrospun PVDF acting as the supporting framework and the 3D cross-linked network structure.

2.6. Galvanostatic Li Plating/Stripping Cycling Tests

Dynamic stability of the Li/electrolyte interface was investigated by galvanostatic Li plating/stripping cycling test of Li|LPD@PVDF|Li symmetric cell and Li|LE-Celgard|Li symmetric cell. The symmetric Li/Li cells were charged/discharged at current density of 0.5 mA cm^{-2}. As shown in **Figure 5**a, overpotential of Li|LE-Celgard|Li symmetric cell rises sharply after cycled about 350 h. After 400 h, it showed unstable dips and bumps with a much larger overpotential. After Li plating/stripping cycling test, the morphology of Li metal electrode surface was analyzed by SEM (Figure 5b). The lithium electrode in contact with the LE-Celgard electrolyte is rough with some stick-shaped grains at surface. So it can be concluded that unstable and high overpotential of Li|LE-Celgard|Li symmetric cell is due to unstable Li/electrolyte interface and dendrite growth. In contrast, Li|LPD@PVDF|Li symmetric cell shows an excellent cycle performance under the same conditions. Steady overpotential is as low as 0.033 V. Even cycled for over 1050 h, the cell maintains an extremely stable and low overpotential without short circuiting. Furthermore, Li electrode shows very smooth and dense surface morphology after Li plating/stripping cycling test (Figure 5c). All these demonstrate ability of LPD@PVDF to promote uniform lithium deposition and restrain Li dendrite growth, thus stabilizing LMBs from the short-circuit problem. According to the combination of the model proposed by Chazalviel[8a] and the model proposed by Monroe and Newman,[9] the ability to restrain Li dendrite growth is due to high lithium-ion transference number and good mechanical strength of LPD@PVDF SIPE. As illustrated in Figure 5e, movement of the anions

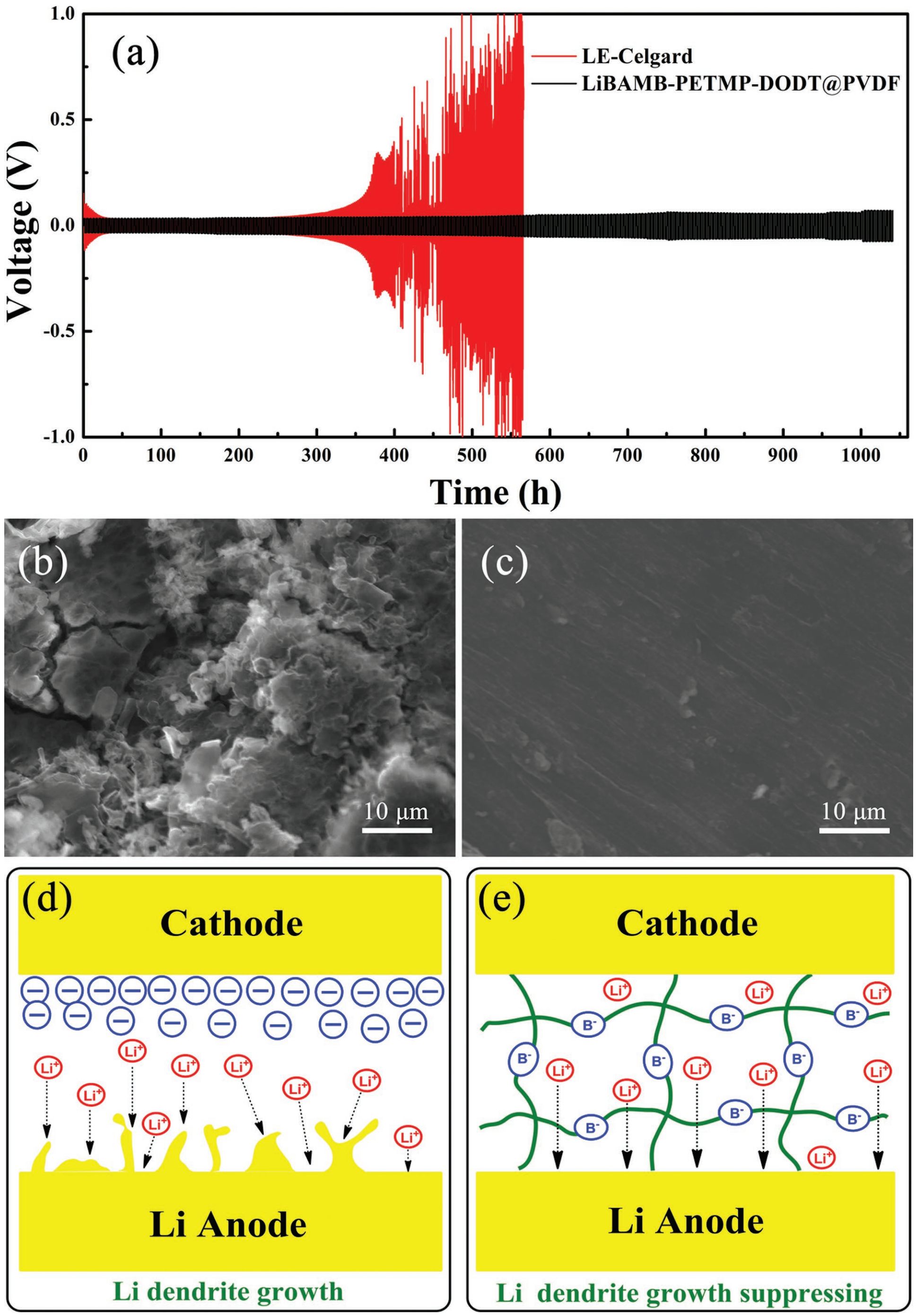

Figure 5. a) Galvanostatic Li plating/stripping cycling profiles of Li|LPD@PVDF|Li symmetric cell and Li|LE-Celgard|Li symmetric cell at density of 0.5 mA cm^{-2}. SEM images of Li metal electrode surface from b) Li|LE-Celgard|Li cell and c) Li|LPD@PVDF|Li cell after Li plating/stripping cycling test. d) Schematic illustration of lithium dendrite growth in LE-Celgard. e) Schematic illustration of LPD@PVDF to suppress lithium dendrite growth.

$BAMB^-$ is restricted by covalently tethered into the polymer network. There is no anion depletion and no electric field caused by anion. Therefore, there is no Li dendrite growth. While in LE-Celgard electrolyte, anions can move freely and the anion depletion near Li electrode could lead to large electric fields, which in turn causes dendrite growth.

2.7. Battery Performance

To evaluate performance of LPD@PVDF SIPEs in lithium metal batteries, LPD@PVDF was used to assemble coin cells, with $LiFePO_4$ as the cathode and lithium foils as the anode. LPD@PVDF SIPEs act as both electrolyte and separator in the Li|LPD@PVDF|$LiFePO_4$ cells. **Figure 6**a shows the rate capability of the cell at various C-rates at 25 °C. Discharge capacity can reach 152 mA h g^{-1} at 0.2 C, which amounts to 90% of the theoretical capacity of $LiFePO_4$. At higher discharge/charge rates, the cell delivers high reversible capacities of 147 mA h g^{-1} at 0.5 C, 140 mA h g^{-1} at 1 C, and 130 mA h g^{-1} at 2 C. It is remarkable that the cell can keep a steady discharge capacity of 112 mA h g^{-1} at 5 C. After 30 cycles, the discharge/charge rate was switched from 5 C back to 0.2 C, and the capacity recovered to 149 mA h g^{-1}. Superior rate capability of the cell is mainly due to the high ionic conductivity and T_+ of the LPD@PVDF SIPEs.

Discharge/charge profiles of the Li|LPD@PVDF|$LiFePO_4$ cell at the discharge/charge rates of 0.2, 0.5, and 1 C are shown Figure 6b. Overpotential value is 0.06 V at discharge/charge rate of 0.2 C, which is far less than 0.11 V of the reported SIPEs.[15h]

Figure 6c shows the long-term discharge/charge cycles at 1 C rate. Li|LPD@PVDF|$LiFePO_4$ cell maintains discharge capacity of 128 mA h g^{-1} at 1 C rate after 380 cycles, which is 91% retention of the initial capacity. In contrast, discharge capacity of the cell with LE-Celgard decays to 103 mA h g^{-1} after 380 cycles (retention rate: 74%). What's more, coulombic efficiency of Li|LPD@PVDF|$LiFePO_4$ cell is higher than that of Li|LE-Celgard|$LiFePO_4$ cell over all cycles. Obviously, the cell with LPD@PVDF SIPE exhibits outstanding cycle performance compared to that of the cell with LE-Celgard, which is attributed to LPD@PVDF SIPE membrane's ability to promote uniform lithium deposition and restrain Li dendrite growth as manifested by galvanostatic Li plating/stripping cycling tests. The result demonstrates that LPD@PVDF SIPE shows promising potential for safe, long-life, and high-rate LMBs applications.

Finally, in order to demonstrate the ability of LPD@PVDF SIPE to be applied to flexible battery, a thin and flexible pouch battery consisting of $LiFePO_4$ (as cathode), lithium metal (as an anode), and LPD@PVDF SIPE (as separator and electrolyte) was fabricated. The battery was able to light up a blue light-emitting diode (LED) lamp and showed a voltage of 3.40 V at the flat states (**Figure 7**a). When the battery was bended 180° (Figure 7b), it can still light up a blue LED lamp and maintained a voltage of 3.40 V. These results demonstrated promising applicability of LPD@PVDF SIPE to flexible batteries for flexible electronic device.

3. Conclusion

In summary, a novel SIPE with a 3D cross-linked network within the electrospun PVDF matrix can be readily prepared. The 3D cross-linked network was constructed by coupling of LiBAMB,

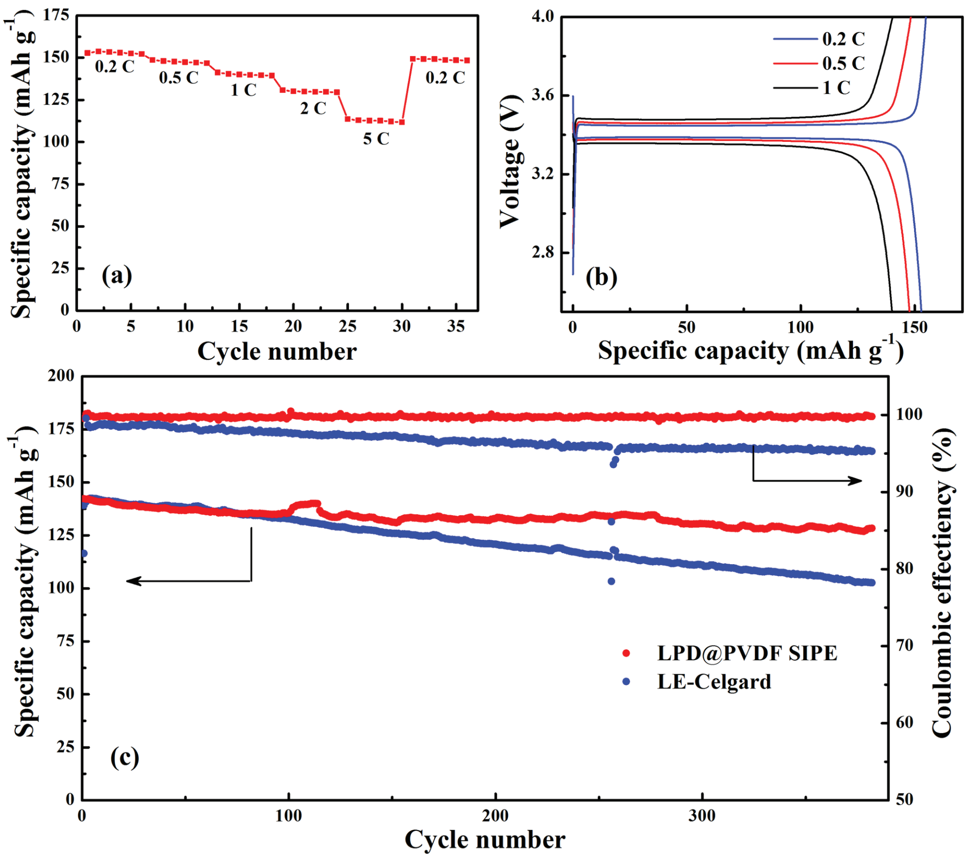

Figure 6. Battery performance of Li|LPD@PVDF|$LiFePO_4$ cells at room temperature: a) rate capability; b) voltage plateaus at different C-rates. c) Discharge capacity and Coulombic efficiency of Li|LPD@PVDF|$LiFePO_4$ cell and Li|LE-Celgard|$LiFePO_4$ cell at discharge/charge rates of 1 C under room temperature.

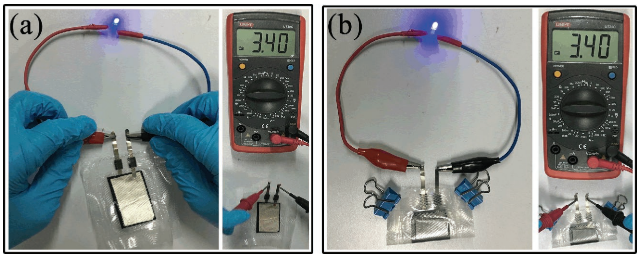

Figure 7. Illustration of Li|LPD@PVDF|$LiFePO_4$ flexible battery: a) at the flat states; b) at the 180° bending states.

PETMP, and DODT via one-step photoinitiated in situ thiol–ene click reaction. It was found that the ionic conductivity of LPD@PVDF SIPE can be optimized by adjusting crosslinking density and precursor concentration. The optimized SIPE showed an outstanding ionic conductivity of 1.32×10^{-3} S cm^{-1} at 25 °C, a high lithium-ion transference number of 0.92, and a wide electrochemical window up to 6.0 V. Owing to the presence of electrospun PVDF matrix, the SIPE exhibited superior mechanical properties with tensile strength of 7.2 MPa and elongation at break of 269%, which can then suppress lithium dendrite growth. The Li|LPD@PVDF|Li symmetric cell maintained an extremely stable and low overpotential without short circuiting over 1050 h cycle. Li|LPD@PVDF|$LiFePO_4$ cell demonstrated excellent rate capacity and cycle performance compared to the cell with LE-Celgard separator. It can keep a steady discharge capacity of 112 mA h g^{-1} at 5 C and maintain discharge capacity of 128 mA h g^{-1} at 1 C rate after 380 cycles, which is 91% retention of the initial capacity. The facile approach of SIPE provides one of the most promising electrolytes for safe, long-life, and high-rate lithium metal batteries.

4. Experimental Section

Materials: 2,2-Dimethoxy-2-phenylacetophenone (DMPA) (Aladdin), GBL (Aladdin), N-methyl-2-pyrrolidone (NMP) (Aladdin), super P (TIMCAL), PVDF (Arkema), and $LiFePO_4$ (MTI Kejing Co., Ltd.) were used as-purchased without further purification. PETMP (Sigma-Aldrich) and DODT (TCI) were dried and stored with 4 Å molecule sieves in a glove box. Other solvents such as acetone and N,N-dimethylacetamide were of analytical reagent grade and used without further purification.

Preparation of Electrospun PVDF Membrane: PVDF was dissolved in a mixed solvent of acetone and N,N-dimethylacetamide (7:3 wt% ratio) to obtain a 16 wt% solution. Electrospinning of the solution was performed at flow rate of 1 mL h^{-1} with a potential difference of 20 kV. Distance between the collector (aluminum drum) and the syringe needle was 16.5 cm. The electrospun PVDF membranes were removed from the collector drum and dried at 60 °C for 12 h. Then the membranes were punched into circular rounds and dried at 50 °C for 24 h under high vacuum.

Preparation Procedure of LPD@PVDF SIPEs: LiBAMB was synthesized following the procedure described in a previous work.[23]

In an argon-filled glove box, LiBAMB (1.8 mmol), PETMP (0.6 mmol), DODT (0.6 mmol), and DMPA (0.18 mmol) were dissolved in GBL (5.4 mL) to form precursor. The precursor was added to electrospun PVDF membranes on a polytetrafluoroethylene (PTFE) plate and excess precursor on the swelled membranes was wiped with air-laid paper. Then it was cured under UV light (365 nm) for 10 min. Prior to use, they were immersed in GBL to reach saturation.

Tensile Strength Measurement: The prepared LPD@PVDF membrane was cut into a rectangular strip with the dimension of 19 mm × 4.5 mm. The tensile strength was investigated using a universal mechanical testing machine (New SANS, Shenzhen, China) under a constant displacement rate of 25 mm min^{-1}.

Ionic Conductivity Measurement: Ionic conductivity of LPD@PVDF SIPEs was measured by EIS using the Solartron 1255B frequency response analyzer. An AC excitation voltage of 10 mV was applied to two stainless steel (SS) electrode cells (loaded with the LPD@PVDF SIPEs) with a frequency range from 1 MHz to 1 Hz. EIS was carried out in a temperature range of 20–80 °C after keeping the samples at each testing temperature for at least 1 h to attain thermal equilibration. The ionic conductivity was calculated using the following equation

$$\sigma = \frac{l}{RA} \tag{1}$$

where l is the film thickness, R stands for the bulk resistance, and A represents the area.

DFT Calculations: All the DFT calculations for the LiBAMB and $Li^+(GBL)_n$ solvation complexes were performed with the GAMESS-US[24] suite of packages using the dispersion-corrected[25] B3LYP (B3LYP-D3).[26] Optimizations and energy calculations were carried out through the 6-31+G(d)[27] basis set followed by single-point energy calculations with the aug-cc-pVTZ basis set.[28] Reported energies are free energies in kcal mol^{-1}. Only vibrational free energy corrections to the electronic energy at 298 K were used in accordance with recommendations for molecules optimized. Normal modes of all structures were examined to verify that equilibrium structures possess no imaginary frequencies.

Lithium-Ion Transference Number Measurement: Lithium-ion transference number was derived from the measurements of CA and EIS before and after the CA test with the cell configuration of "Li/membrane/Li," that is, LPD@PVDF membrane was placed between two pieces of lithium foils and sealed in a 2032 coin cell in a glove box. The cell was placed in a 50 °C oven to carry out measurement. EIS was performed from 1 MHz to 1 Hz with an AC excitation voltage of 10 mV. CA was carried out at a static potential of 10 mV.

The lithium-ion transference number (T_+) was calculated using the following equation[21]

$$T_+ = \frac{I_s(\Delta V - I_0 R_0)}{I_0(\Delta V - I_s R_s)} \tag{2}$$

where ΔV is the potential applied across the cell, I_0 and I_S are the initial current and steady state current, and R_0 and R_S represent the initial resistance and the steady-state resistance of the passivation layers on the lithium electrode.

Electrochemical Stability Window: LPD@PVDF membrane was sandwiched between a SS electrode and a lithium foil to carry out linear scanning voltammetry from 0 to 6.5 V (vs Li^+/Li) at a scan rate of 1 mV s^{-1}.

Battery Assembly and Measurement: $LiFePO_4$, super P, and PVDF were mixed in a 80:10:10 weight ratio in NMP. The mixture was magnetic stirred over 12 h to form a well-dispersed slurry. The flat aluminum foil was coated with the viscous slurry by the doctor blade process. The obtained electrode films were dried at 60 °C for 30 min in an air-circulating oven and further dried at 50 °C under high vacuum for 24 h. The coin cell (CR2032 type) was made by sandwiching LPD@PVDF membrane between a lithium metal foil and a $LiFePO_4$ electrode in an argon-filled glove box (H_2O and O_2 content <0.1 ppm). Charge–discharge tests were carried out with the upper cut-off voltage of 4.0 V and the lower cut-off voltage of 2.5 V at room temperature.

Galvanostatic Li Plating/Stripping Cycling Tests: Galvanostatic charge/discharge test was carried out to evaluate dendrite-suppressing behavior. Symmetric Li/Li cells were cycled under constant current density of 0.5 mA cm^{-2} and the polarity was reversed for every 1 h.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

flexible, lithium dendrites, lithium-ion transference numbers, lithium metal batteries, single-ion conducting electrolytes

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